

Airbnb Price Prediction Using Linear Regression

In this project, I built a machine learning model to predict Airbnb house prices based on different property features. The main goal was to understand how factors like room type, number of bedrooms, bathrooms, reviews, and availability affect the price of an Airbnb listing.

First, I selected useful columns from the dataset such as price, accommodates, bathrooms, bedrooms, beds, room type, property type, minimum nights, number of reviews, review score rating, availability, and instant booking option. Since some values were missing in the dataset, I handled them by filling numerical missing values with the median value so that the data could be used properly for model training.

Next, categorical columns like room type, property type, and instant booking were converted into numerical form using One-Hot Encoding, because machine learning models cannot directly work with text values.

After preprocessing, I split the dataset into training and testing parts. 80% of the data was used to train the model and 20% was used for testing. Then I created a pipeline combining preprocessing and Linear Regression so that all steps run together smoothly.

After training the model, I checked performance using evaluation metrics. The model gave MAE: 31.00, RMSE: 59.60, and R^2 Score: 0.63. This means the model predicts prices with a reasonable average error, and about 63% of the price variation is explained by the model.

To understand the results visually, I plotted actual prices against predicted prices. Most points were close to the diagonal line, which shows that the model predicts many listings correctly. Some expensive listings were harder to predict accurately, which is common in Linear Regression.

Overall, this project helped me understand the full machine learning workflow including data cleaning, preprocessing, model building, prediction, evaluation, and visualization.